PURDUE UNIVERSITY

Course Information/Description

- **AAE 33400: Aerodynamics**
  Incompressible airfoil and lifting line theory. Steady and unsteady, one-dimensional, linear and nonlinear flows. Normal and oblique shock waves. Isentropic expansions. Steady, supersonic, two-dimensional linear and nonlinear flow applied to airfoils. Linear subsonic compressible flow for airfoils. Design applications.
- **Instructional Modality:** Face-to-Face
- **Course credit hours:** 3
- **Prerequisites:** AAE 33300 (Fluid Mechanics), AAE 33301 (Fluid Mechanics Lab), ME 20000 (Thermodynamics). ME 30800 (Introductory Fluid Mechanics) may serve in place of AAE 33300 and AAE 33301.

Learning Resources & Texts

- **Required Text:** J. D. Anderson "Fundamentals of Aerodynamics," 7th Edition, McGraw Hill, 2024 (ISBN: 9781264151929). (While the 7th edition is current, any edition of the book is acceptable, although older editions do not have the compressible flow tables or standard atmosphere table in the appendices. More recent editions also have added material related to design.)

  The textbook is a useful resource for gaining a deeper understanding of the material presented in class. Also, the textbook can provide a second explanation for how to approach a problem. Unlike in AAE 333, reading quizzes will not be given, but students who read the book before coming to class will find they gain a deeper understanding of the material.

- **Selected Useful References:**
  - E.L. Houghton, P.W. Carpenter, S.H. Collicott, and D.T. Valentine, *Aerodynamics for Engineering Students, 7th Edition,* Elsevier, 2016.
  - A. M. Kuethe and C.Y. Chow, *Foundations of Aerodynamics*, Wiley, 5th ed. 1998.
  - J.D. Anderson, *Modern Compressible Flow*, McGraw Hill, 3rd ed. 2003.
  - J. Katz and A. Plotkin, *Low-Speed Aerodynamics*, Cambridge U. Press, 2nd ed. 2001
  - J. Moran, *An Introduction to Theoretical and Computational Aerodynamics*, 2nd edition, 2003.

  These references are books are not required but can provide additional insight into the topics covered in class.

- **Use of artificial intelligence (AI) or Large Language Models (LLM) in this course**

  It seems likely that the use of AI and LLM will increase over the next several years. Therefore, it seems prudent to allow students to use AI while in school when appropriate. However, we must balance the use of AI with the desire for you to obtain knowledge and have the ability to identify when AI has "hallucinated". Also, we acknowledge that copyright should be upheld and that we should not share the work of others with AI without their permission. For a thought-provoking discussion of the affects of AI on learning, see the article, "The Memory Paradox: Why Our Brains Need Knowledge in an Age of AI," posted in the Supplementary Material folder on Brightspace. With these ideas in mind, the following policies will be enforced in this class:

  - **AI Policy 1: You may not share course materials with AI.** Class recordings, notes, and handouts are considered to be 'derivative works' of the instructor's presentations and materials, and they are thus subject to the instructor's (or original content producer's) copyright. This also includes homework assignments, exams and solutions. As such, they should not be entered into any AI system or uploaded to any file sharing site, sold or bartered without express written permission. To be clear: this means that uploading any course material on "CourseHero", ChatGPT, Copilot (even the Purdue branded version) or similar websites is a violation of the course policy and a violation of copyright laws.
  - **AI Policy 2: You may use AI to help you on Homework but not on the Exams.** This policy attempts to balance learning how to use AI while gaining knowledge in the area of aerodynamics. Therefore, you are allowed to use AI while you are developing your knowledge in the field (on formative assessments), but not when your personal knowledge is assessed (on summative assessments). Note that from AI Policy 1, you may not copy and paste homework problem statements into AI systems, since that violates copyright laws. You may formulate your own questions to ask AI that will help you in doing the homework problem. However, keep in mind that, even if AI gives you a numerical answer for some quantity to be computed, in most cases to receive any credit for a homework problem you must show your work, including how that number is computed from the relevant equations. (Use of AI can be considered a form of collaboration. See the policy statement on **Collaboration**.)
  - **AI Policy 3: AI use must be tracked and acknowledged**. If you used Generative AI programs such as ChatGPT or CoPilot to assist with your assignment, you must acknowledge its use by following the guidelines provided in Monash University's Acknowledging the Use of Generative Artificial Intelligence. Specify how and where your readers can expect to see the impact, and include an Appendix for the assignment that shows what aspects were generated. ChatGPT now includes the ability to share links to conversations; you can use extensions such as ShareGPT to share your ChatGPT conversations, or you can include screenshots. (This paragraph was adopted from Dr. Kirsten Davis, from the ENE 50101 course syllabus.)

- **AI Policy 4: You may not use paid versions of AI.** Paid versions of AI are often more powerful and not all students can afford to use the paid versions of AI tools, potentially placing some students at a disadvantage. Therefore, only AI tools that are freely available to all Purdue students may be used in this class.

- The Helen Bass Williams Academic Success Center, provides a variety of proactive, practical and approachable academic support services for undergraduate students in West Lafayette and Indianapolis. See the Programs & Services tab on the ASC website for details

- The Purdue Libraries and School of Information Studies support faculty and students in West Lafayette and Indianapolis with many other resources. Check out the recently redesigned website. You may reach out via the Ask a Librarian virtual chat service, email or by scheduling a consultation with a librarian.

- Brightspace Learning Management System:

  Access the course videos, materials, and resources via Purdue's BS learning management system. Begin with the Start Here tab, which describes how the course Brightspace is organized. We strongly suggest that you explore and become familiar not only with the site navigation but with content and resources available for this course. See the Student Services widget on the campus homepage for resources such as Technology Help, Academic Help, Campus Resources, and Protect Purdue.

  The lecture notes will be made available on Brightspace. Most of the lecture notes have blank spaces where information will be added during class. Students are responsible for printing or downloading the course notes before lecture so they can add material to the notes during class. If you miss class, please get the missing material for the notes from another student.

  The course notes contain copyrighted material. Therefore, do not post the files on the Internet or share them with anyone not enrolled in this course. This includes a prohibition against uploading course material to AI systems (see AI Policy 1).

Learning Outcomes

On completing this course, you will be able to:

- Calculate thin airfoil performance parameters for incompressible flow
- Calculate general wing loading by lifting line theory and compare to elliptic loading case
- Determine horizontal tail location to maintain trim and a specified static margin
- Compute isentropic stagnation conditions and apply in problem solving
- Compute jumps in properties across steady shocks and expansions

- Determine supersonic airfoil performance by shock-expansion method
- Determine supersonic airfoil performance by linearized supersonic theory
- Apply subsonic compressibility corrections to incompressible results
- Determine critical Mach number for sub-sonic flight
- Determine supply or back pressures for supersonic nozzle operating conditions

When you take all the learning outcomes together, you should be able to calculate the lift of an airfoil and wing over a range of Mach numbers and compute the conditions inside of a converging-diverging nozzle like that on a rocket.

Assessments: Homework Assignments and Exams

Your achievement of the course learning outcomes will be assessed using homework and exams. Homework will account for 25% of your grade and exams will count for 75%.

Assessments	Due	Percentage
Homework	Weekly, throughout the semester	25
Exam 1	TBD	25
Exam 2	TBD	25
Final Exam	TBD	25
		Total: 100

Grading Scale

Course grades will be <u>no more severe</u> than:

97.0-100% :	A+	77.0-79.99%:	C+
93.0-96.99%:	A	73.0-76.99%:	C
90.0-92.99%:	A-	70.0-72.99%:	C-
87.0-89.99%:	B+	67.0-69.99%:	D+
83.0-86.99%:	B	63.0-66.99%:	D
80.0-82.99%:	B-	60.0-62.99%:	D-
		<60.0%:	F

Final grades may be curved or offset to a lower scale than what is shown above.

Note on rounding: Rounding is not done. Once the grading scale is finalized grades are set and no adjustments will be made.